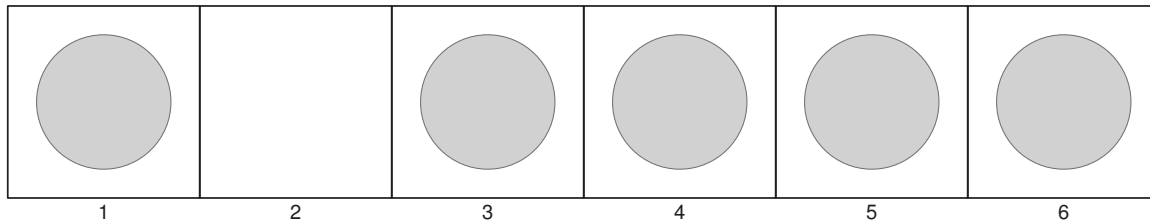


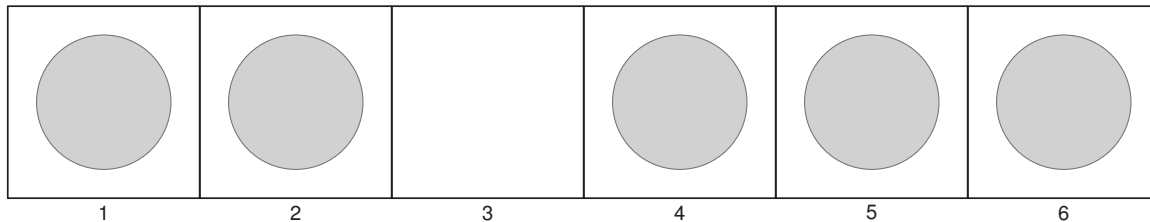
**Problem 1:** A penny may jump over the penny next to it, to the left or to the right, into the empty square right behind it. The penny that was jumped over comes off the strip. Strips A, B and C have six squares each, with the empty square in a different place on each. On each strip, put a penny on each circle and play until no jump is left, trying to leave as few pennies as you can. Play each strip at least three times, making different choices where you can. Draw every different ending you find on the small strips under it and write its number of pennies. On strip A, circle the square where the last penny sat in each ending. On strip B, write beside each ending the number of the square its first jump started from. Write the fewest pennies you could leave on each strip in the number box at the right.

**A**


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**B**


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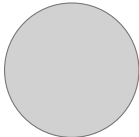
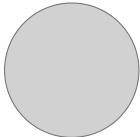
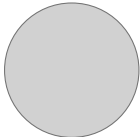
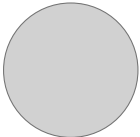
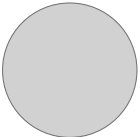
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**C**

|                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                    |   |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|---|
|  |  |  |  |  |   |
| 1                                                                                 | 2                                                                                 | 3                                                                                 | 4                                                                                 | 5                                                                                  | 6 |



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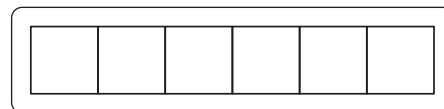
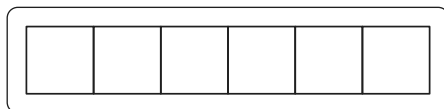
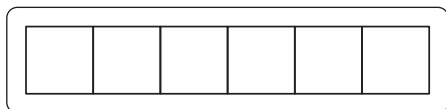
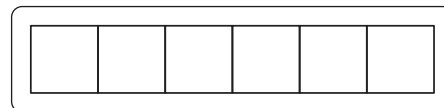
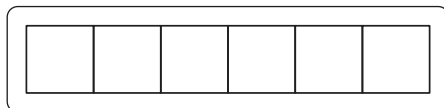
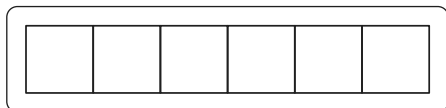
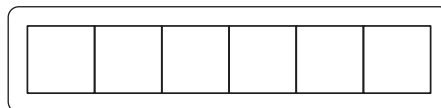
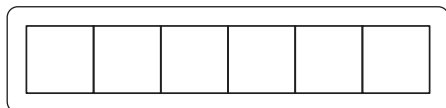
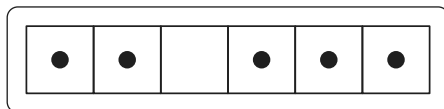
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**Problem 2:** Make a map of every position you can reach on strip B. The top box shows the start. In the boxes below, draw every position that one jump can reach from the start, with an arrow from the start to each one. From each new position, draw every position that one more jump can reach, with an arrow. Stop when every position is either already on the map or has no jump left. Circle the positions with no jump left and write the number of pennies in each. Count the positions on the map and write the number in the number box.



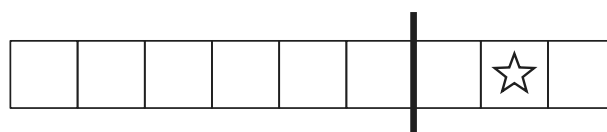
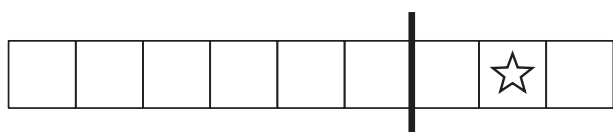
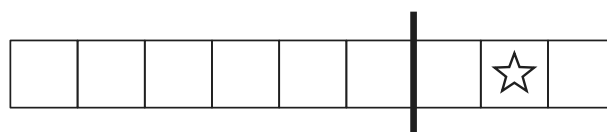
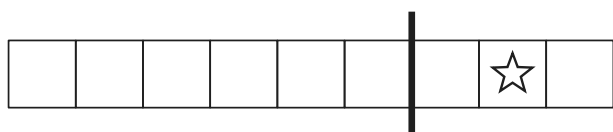
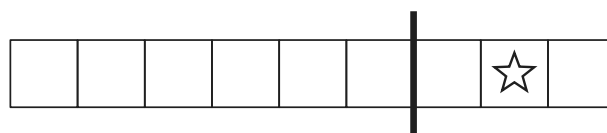
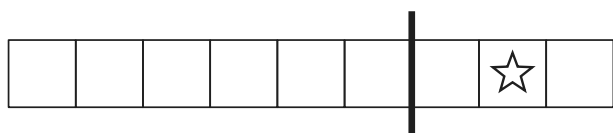
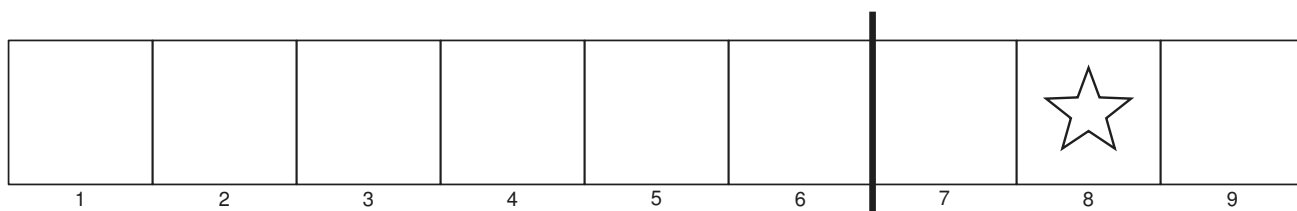
**Problem 3:** Strips D, E and F have a thick line across them. The pennies start on the left of the line, on the circles. Jump so that a penny gets as far to the right as it can. Play each strip twice. On the small strip under each, put a cross on the farthest square any penny reached, and write how many squares past the line it is. Strip G has nine squares, a line after square 6, and a star on square 8. Put pennies on any of the squares 1 to 6, as many or as few as you like, and try to get a penny onto the star. Draw each army you try as dots on a small strip and put a cross on the farthest square a penny reached. Try at least six different armies, some with gaps.

**D**

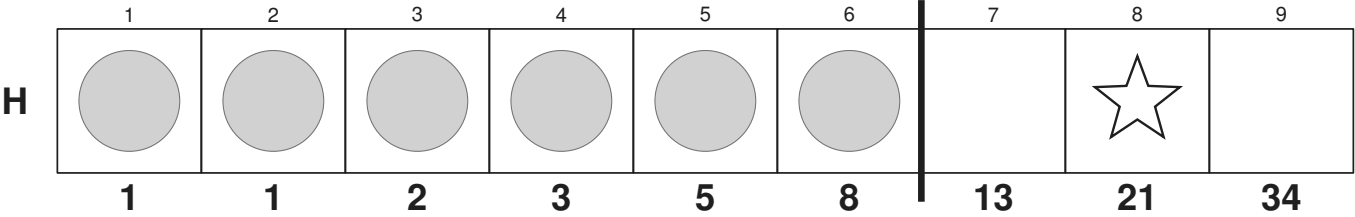
**E**

**F**

G



**Problem 4:** Strip H is strip G with a price printed under every square: 1, 1, 2, 3, 5, 8 on squares 1 to 6, then 13, 21, 34 past the line. Put a penny on each of the squares 1 to 6. Add up the prices of the squares your pennies are on and write the total in the first row of table 1. Make a jump, add up the prices of the squares that have pennies now, and write the new total in the next row. Continue until no jump is left, writing the total after every jump. Play strip H a second time with different jumps and fill in table 2. Then do the same on strip J, where the pennies start on squares 1, 2, 4, 5 and 6, and fill in table 3. Write under the tables what a jump to the right does to the total and what a jump to the left does to it. Write the price of the star square beside the largest total in your tables. Last, fill in the small table: an army filling squares 1 to 3 is worth  $1 + 1 + 2 = 4$ , and the square two past it, square 5, costs 5. Do the rows for armies filling squares 1 to 4, 1 to 5, 1 to 6 and 1 to 7, and write the difference in the last column.

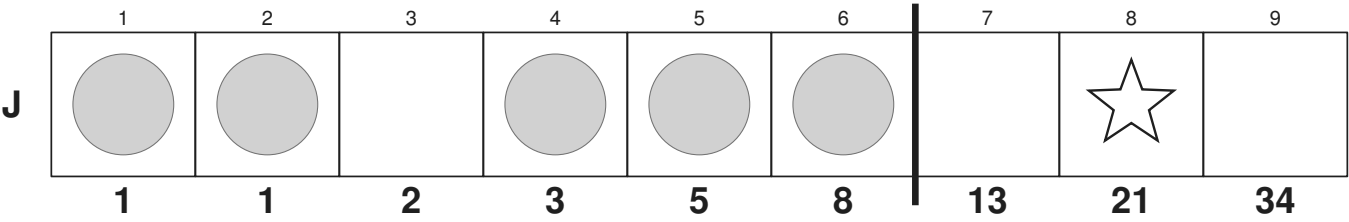


**1**

| after jump | total |
|------------|-------|
| 0          |       |
| 1          |       |
| 2          |       |
| 3          |       |
| 4          |       |
| 5          |       |

**2**

| after jump | total |
|------------|-------|
| 0          |       |
| 1          |       |
| 2          |       |
| 3          |       |
| 4          |       |
| 5          |       |

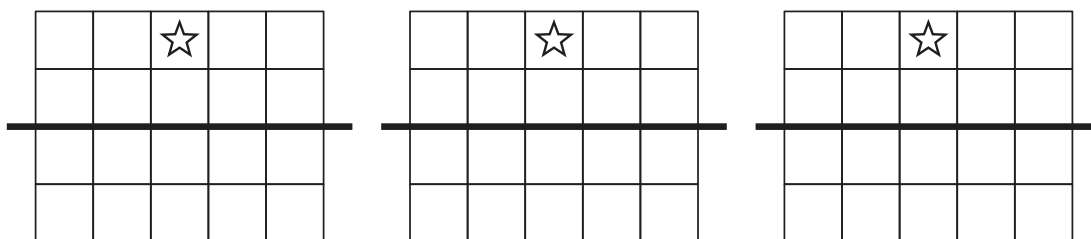
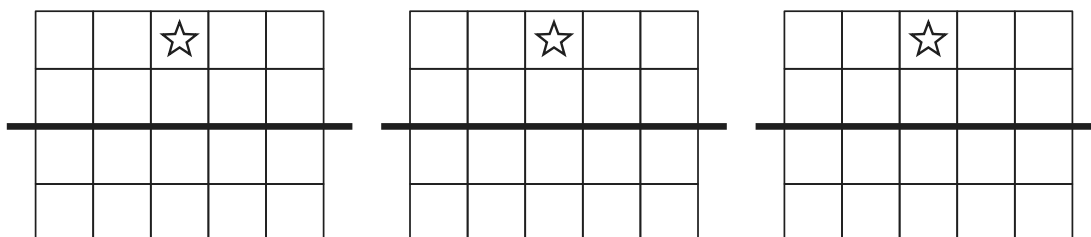
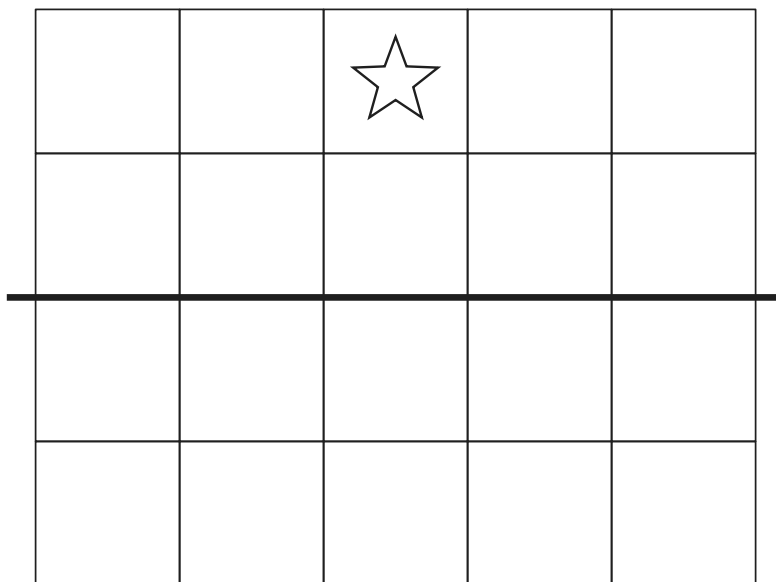


3

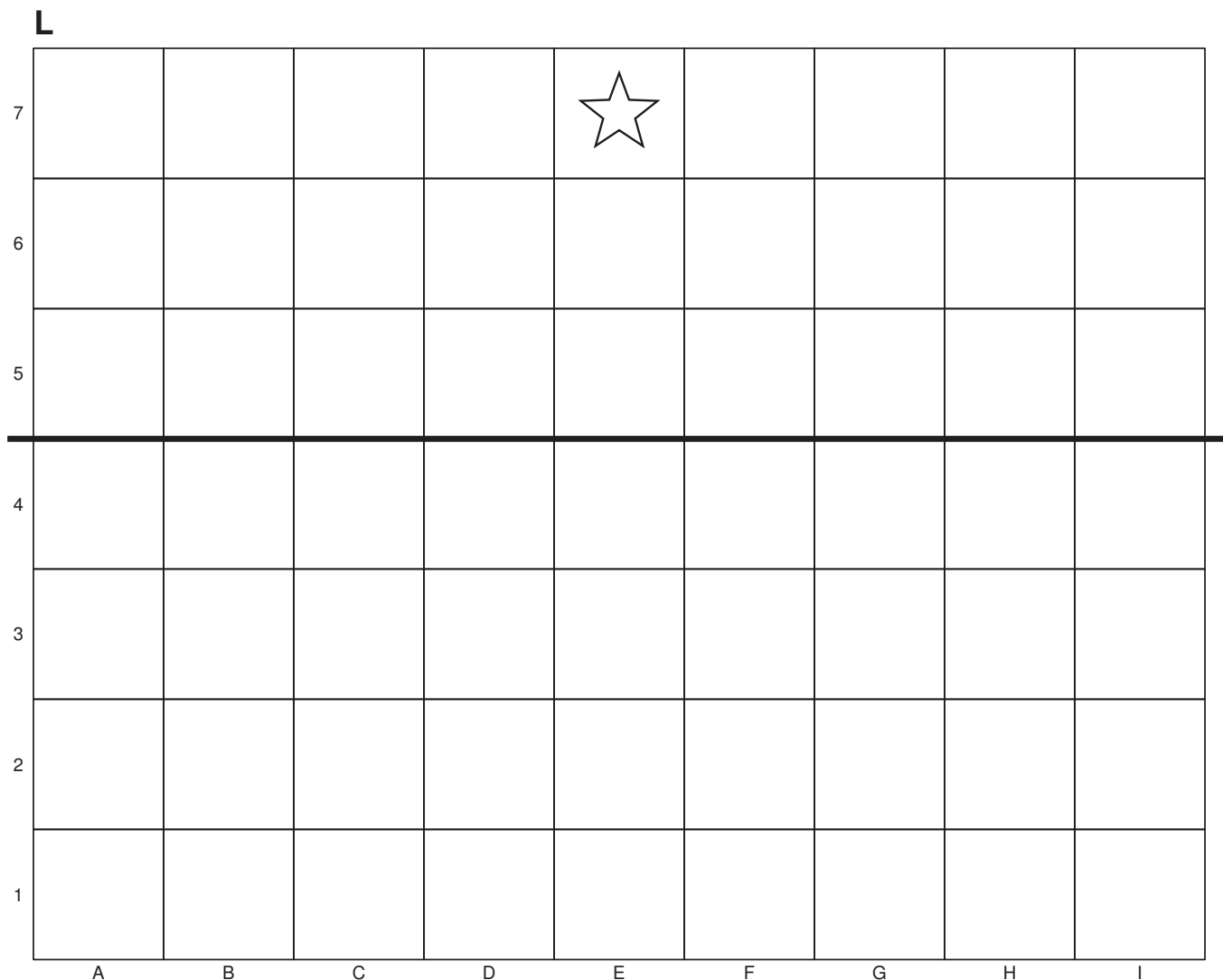
| after jump | total |
|------------|-------|
| 0          |       |
| 1          |       |
| 2          |       |
| 3          |       |
| 4          |       |
| 5          |       |

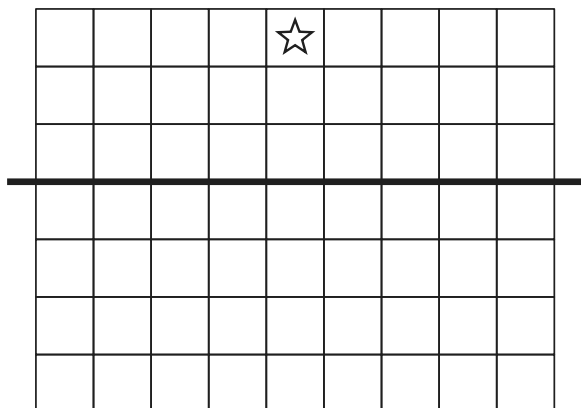
| army fills squares | army's worth | price of the square two past | difference |
|--------------------|--------------|------------------------------|------------|
| 1 to 3             | 4            | 5                            | 1          |
| 1 to 4             |              |                              |            |
| 1 to 5             |              |                              |            |
| 1 to 6             |              |                              |            |
| 1 to 7             |              |                              |            |

**Problem 5:** On board K the star is two rows above the line. Pennies may also jump up or down, over the penny next to them into the empty square right behind it. Take exactly three pennies, put them anywhere below the line, and try to land one on the star. Draw each army of three you tried on a small board. Then take exactly four pennies and find a placement that works. Draw every army of four that works on a small board, with its jumps drawn as arrows numbered 1, 2, 3.

**K**

**Problem 6:** On board L the star is three rows above the line. Land a penny on the star using exactly eight pennies, placed anywhere below the line. Draw the army you used on small board 1, and list its seven jumps in order in the blanks, naming each square by its column letter and row number, for example C2 over C3 to C4. Then find a second army of eight that works and is not a mirror image of the first. Draw it on small board 2 and list its jumps.



**1**

1. \_\_\_\_\_ over \_\_\_\_\_ to \_\_\_\_\_

2. \_\_\_\_\_ over \_\_\_\_\_ to \_\_\_\_\_

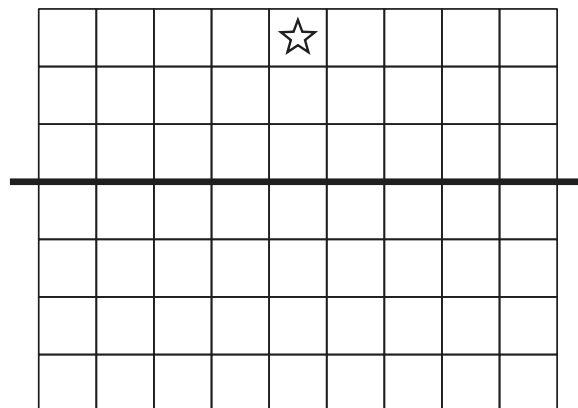
3. \_\_\_\_\_ over \_\_\_\_\_ to \_\_\_\_\_

4. \_\_\_\_\_ over \_\_\_\_\_ to \_\_\_\_\_

5. \_\_\_\_\_ over \_\_\_\_\_ to \_\_\_\_\_

6. \_\_\_\_\_ over \_\_\_\_\_ to \_\_\_\_\_

7. \_\_\_\_\_ over \_\_\_\_\_ to \_\_\_\_\_

**2**

1. \_\_\_\_\_ over \_\_\_\_\_ to \_\_\_\_\_

2. \_\_\_\_\_ over \_\_\_\_\_ to \_\_\_\_\_

3. \_\_\_\_\_ over \_\_\_\_\_ to \_\_\_\_\_

4. \_\_\_\_\_ over \_\_\_\_\_ to \_\_\_\_\_

5. \_\_\_\_\_ over \_\_\_\_\_ to \_\_\_\_\_

6. \_\_\_\_\_ over \_\_\_\_\_ to \_\_\_\_\_

7. \_\_\_\_\_ over \_\_\_\_\_ to \_\_\_\_\_

**Problem 7:** Board M is board L with a price printed in every square. Put a penny on each of the eight circles. Add up the prices of the eight squares and write the total in the first row of the table. Make the seven jumps that land a penny on the star, and after each jump write the total price of the squares that have pennies. Then put the pennies back on the circles, make one jump that moves a penny away from the star, and write the new total in the last row. Next, find the seven most expensive squares below the line on board M, add up their prices, and write the sum beside the star's price. On board N the star is two rows above the line. Find the three most expensive squares below its line, add up their prices, and write the sum beside that star's price.

**M**

|   |   |   |   |   |    |   |   |   |   |
|---|---|---|---|---|----|---|---|---|---|
| 7 | 2 | 3 | 5 | 8 | 13 | 8 | 5 | 3 | 2 |
| 6 | 1 | 2 | 3 | 5 | 8  | 5 | 3 | 2 | 1 |
| 5 | 1 | 1 | 2 | 3 | 5  | 3 | 2 | 1 | 1 |
| 4 | 1 | 1 | 1 | 2 | 3  | 2 | 1 | 1 | 1 |
| 3 | 1 | 1 | 1 | 1 | 2  | 1 | 1 | 1 | 1 |
| 2 | 1 | 1 | 1 | 1 | 1  | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1  | 1 | 1 | 1 | 1 |
|   | A | B | C | D | E  | F | G | H | I |

| after jump | total |
|------------|-------|
| 0          |       |
| 1          |       |
| 2          |       |
| 3          |       |
| 4          |       |
| 5          |       |
| 6          |       |
| 7          |       |
| away       |       |

|   |   |   |   |   |    |   |   |   |   |
|---|---|---|---|---|----|---|---|---|---|
| 7 | 2 | 3 | 5 | 8 | 13 | 8 | 5 | 3 | 2 |
| 6 | 1 | 2 | 3 | 5 | 8  | 5 | 3 | 2 | 1 |
| 5 | 1 | 1 | 2 | 3 | 5  | 3 | 2 | 1 | 1 |
| 4 | 1 | 1 | 1 | 2 | 3  | 2 | 1 | 1 | 1 |
| 3 | 1 | 1 | 1 | 1 | 2  | 1 | 1 | 1 | 1 |
| 2 | 1 | 1 | 1 | 1 | 1  | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1  | 1 | 1 | 1 | 1 |
|   | A | B | C | D | E  | F | G | H | I |

13 \_\_\_\_\_

**N**

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 2 | 3 | 5 | 8 | 5 | 3 | 2 |
| 1 | 2 | 3 | 5 | 3 | 2 | 1 |
| 1 | 1 | 2 | 3 | 2 | 1 | 1 |
| 1 | 1 | 1 | 2 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 |

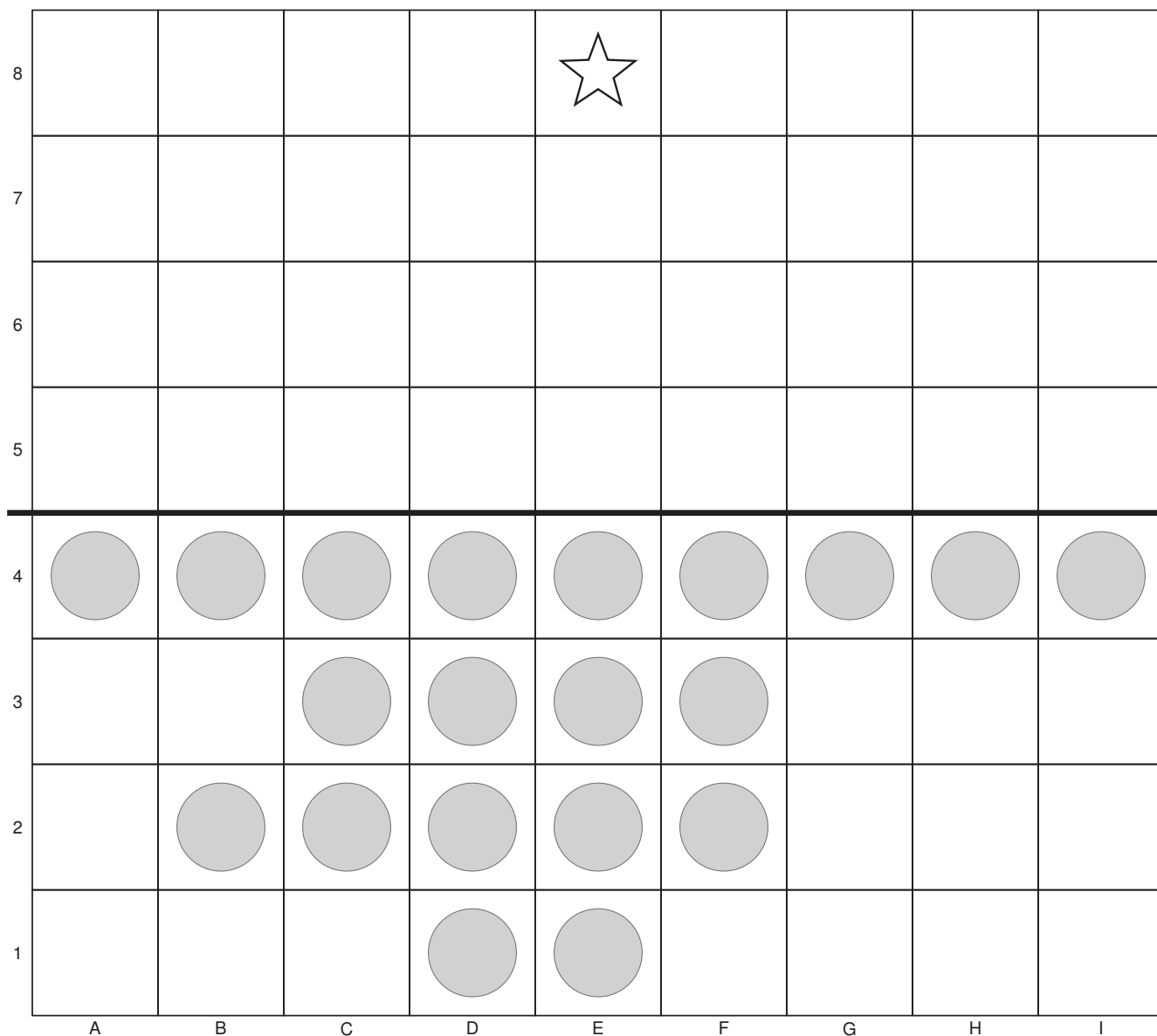
8 \_\_\_\_\_

**Problem 8:** Fill in the table with the fewest pennies that can land a penny one row, two rows and three rows above the line, using what you found in Problems 5 and 6 and at the start of the session. Write your guess for four rows and for five rows. On board P the star is four rows above the line. Try to land a penny on the star with 16 pennies placed anywhere below the line. Write in the number box the highest row above the line that any penny reached. Then use board Q. Put a penny on each of the 20 circles and find the 19 jumps that land a penny on the star. List them in order in the blanks by square names.

| 1 row | 2 rows | 3 rows | 4 rows | 5 rows |
|-------|--------|--------|--------|--------|
|       |        |        |        |        |

**P**

|  |  |  |  |                                                                                   |  |  |  |  |
|--|--|--|--|-----------------------------------------------------------------------------------|--|--|--|--|
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**Q**

1. \_\_\_\_ over \_\_\_\_ to \_\_\_\_

2. \_\_\_\_ over \_\_\_\_ to \_\_\_\_

3. \_\_\_\_ over \_\_\_\_ to \_\_\_\_

4. \_\_\_\_ over \_\_\_\_ to \_\_\_\_

5. \_\_\_\_ over \_\_\_\_ to \_\_\_\_

6. \_\_\_\_ over \_\_\_\_ to \_\_\_\_

7. \_\_\_\_ over \_\_\_\_ to \_\_\_\_

8. \_\_\_\_ over \_\_\_\_ to \_\_\_\_

9. \_\_\_\_ over \_\_\_\_ to \_\_\_\_

10. \_\_\_\_ over \_\_\_\_ to \_\_\_\_

11. \_\_\_\_ over \_\_\_\_ to \_\_\_\_

12. \_\_\_\_ over \_\_\_\_ to \_\_\_\_

13. \_\_\_\_ over \_\_\_\_ to \_\_\_\_

14. \_\_\_\_ over \_\_\_\_ to \_\_\_\_

15. \_\_\_\_ over \_\_\_\_ to \_\_\_\_

16. \_\_\_\_ over \_\_\_\_ to \_\_\_\_

17. \_\_\_\_ over \_\_\_\_ to \_\_\_\_

18. \_\_\_\_ over \_\_\_\_ to \_\_\_\_

19. \_\_\_\_ over \_\_\_\_ to \_\_\_\_